2 A rod PQ is attached at P to a vertical wall, as shown in Fig. 2.1.

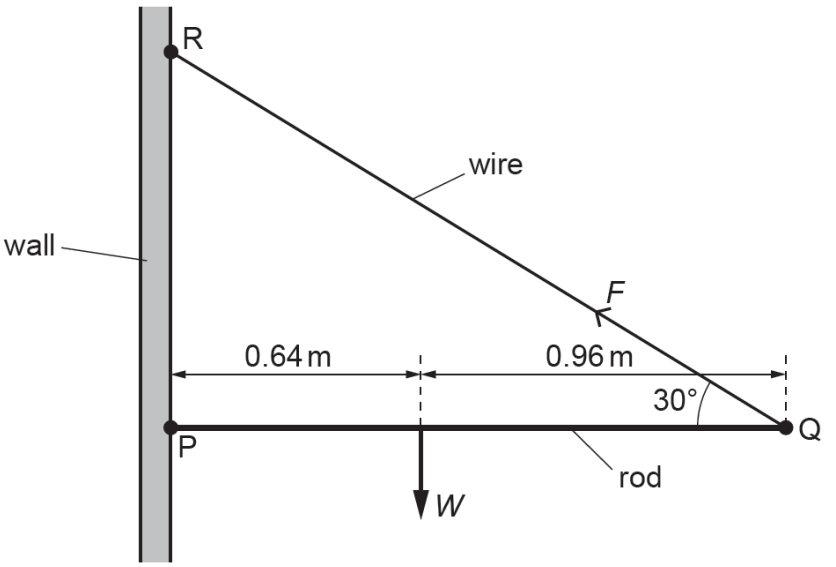

Fig. 2.1

The length of the rod is 1.60 m. The weight *W* of the rod acts at 0.64 m from P. The rod is kept horizontal and in equilibrium by a light wire attached to Q and to the wall at R. The wire provides a force *F* of 44 N on the rod at 30° to the horizontal.

(a) Determine

(i) the vertical component of *F,*

vertical component = ... N [1]

(ii) the horizontal component of *F.*

horizontal component = ... N [1]

(b) Determine the weight *W* of the rod.

W = ... N [2]

(c) Explain why the wall must exert a force on the rod at P to keep the rod in equilibrium.

.....

.....

.....

..... [2]

(d) On Fig. 2.1, draw an arrow to represent the force acting on the rod at P. Label your arrow with the letter S. Explain how you arrive at the answer.

.....

.....

..... [2]

(e) Fig. 2.2 and Fig. 2.3 show two set-ups where the wire is attached to a different point on the rod.

Draw an arrow on each figure to represent the force acting on the rod at P. Label your arrows with the letter S_1 and S_2 respectively.

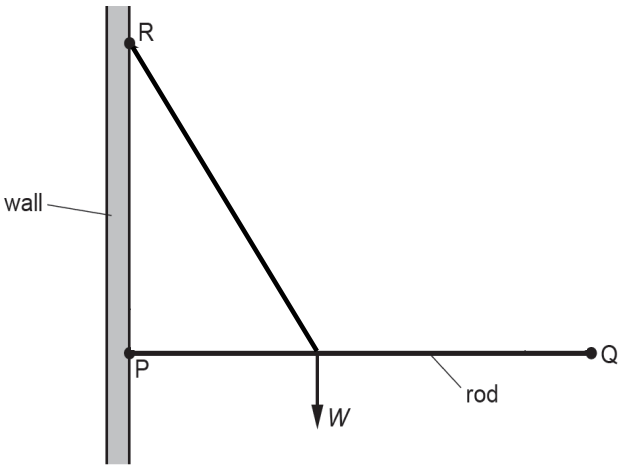

Fig. 2.2

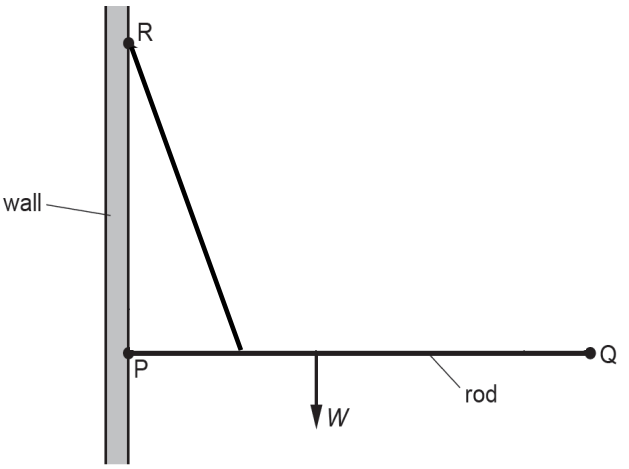

Fig. 2.3

[2]